

Practice item 13

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User: Diya Jain
In this exercise, you need to compute 107 + 9 using 8 - bit 1's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary
Please convert 107 to signed decimal to 8-bit binary representation with 1's complement: 01101011
Please convert 9 to signed decimal to 8-bit binary representation with 1's complement: 00001001

Step 2a: Perform binary addition
      0 1 1 0 1 0 1 1
    + 0 0 0 0 1 0 0 1
    -----
Please input the final carry bit (0 or 1)? 0
Please input the above binary addition result: 01110100
      0 1 1 0 1 0 1 1
    + 0 0 0 0 1 0 0 1
    -----
      0 1 1 1 0 1 0 0

Step 2b: End-Round Carry?
End-round carry operation is required for this binary addition? (y/n): n

Step 3: Verify the result.
Please convert the result binary sum to decimal value: 116
Overflow occurs or not (y/n)? n

The final result is correct and no overflow occurs.
Well done!
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User: Diya Jain
In this exercise, you need to compute 14 + 22 using 8 - bit 1's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary
Please convert 14 to signed decimal to 8-bit binary representation with 1's complement: 00001110
Please convert 22 to signed decimal to 8-bit binary representation with 1's complement: 00010110

Step 2a: Perform binary addition
      0 0 0 0 1 1 1 0
    + 0 0 0 1 0 1 1 0
    -----
Please input the final carry bit (0 or 1)? 0
Please input the above binary addition result: 00100100
      0 0 0 0 1 1 1 0
    + 0 0 0 1 0 1 1 0
    -----
      0 0 1 0 0 1 0 0

Step 2b: End-Round Carry?
End-round carry operation is required for this binary addition? (y/n): n

Step 3: Verify the result.
Please convert the result binary sum to decimal value: 36
Overflow occurs or not (y/n)? n

The final result is correct and no overflow occurs.
Well done!
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Practice item 14

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Please select the practice item: 14

User: Diya Jain
In this exercise, you need to compute 2 + 116 using 8 - bit 2's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary
Please convert 2 to signed decimal to 8-bit binary representation with 2's complement: 00000010
Please convert 116 to signed decimal to 8-bit binary representation with 2's complement: 01110100

Step 2: Perform binary addition
      0 0 0 0 0 0 1 0
    + 0 1 1 1 0 1 0 0
    -----
Please input the final carry bit (0 or 1)?: 0
Please input the above binary addition result: 01110110
      0 0 0 0 0 0 1 0
    + 0 1 1 1 0 1 0 0
    -----
      0 1 1 1 0 1 1 0

Step 3: Verify the result.
Please convert the result binary sum to decimal value: 118
Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.
Well done!
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Please select the practice item: 14

User: Diya Jain
In this exercise, you need to compute 77 + 25 using 8 - bit 2's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary
Please convert 77 to signed decimal to 8-bit binary representation with 2's complement: 01001101
Please convert 25 to signed decimal to 8-bit binary representation with 2's complement: 00011001

Step 2: Perform binary addition
      0 1 0 0 1 1 0 1
    + 0 0 0 1 1 0 0 1
    -----
Please input the final carry bit (0 or 1)?: 0
Please input the above binary addition result: 01100110
      0 1 0 0 1 1 0 1
    + 0 0 0 1 1 0 0 1
    -----
      0 1 1 0 0 1 1 0

Step 3: Verify the result.
Please convert the result binary sum to decimal value: 102
Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.
Well done!
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